

PRASHANT BHANDARI

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SUMMARY

Electronics Engineering student specializing in PCB design, embedded firmware development, and IoT/robotics systems. Proven track record with ESP32, KiCad, C/C++, and MicroPython; shipped hardware from schematic through production across satellite, industrial, and consumer IoT domains. International competition finalist with hands-on leadership of 350+ students in embedded systems.

TECHNICAL SKILLS

PCB Design: KiCad, Proteus, Schematic Capture, PCB Layout, Buck/Boost Converter Design, DRC/ERC

Embedded Systems: ESP32, Arduino, FreeRTOS, ECU2000, HCM111Z, BLE, LoRa, UART/SPI/I2C

Programming: C/C++, Python, MicroPython, JavaScript

Hardware: IoT Systems, Power Electronics, QC Test Fixtures, Robotics, CubeSat Subsystems

Tools: Git, OLED Dashboards, Sensor Integration, OTA, JLCPCB/Gerber Workflows

EXPERIENCE

Electronics Design Intern

Sept 2025 – Jan 2026

SEDS Nepal (Remote)

Pokhara, Nepal

- Designed electronics modules for pseudo-satellite (HAPS) systems, ensuring modular and reproducible architecture.
- Performed component selection, BOM budgeting, and design reviews across interdisciplinary satellite subsystems.

Electronics Engineer Intern

Jan 2025 – Jul 2025

YarsaTech

Pokhara, Nepal

- Designed and optimized buck converter circuits; improved layout across multiple iterations using KiCad.
- Developed automated QC testing device for power backup products using MicroPython and custom firmware.
- Built BLE-enabled QR/barcode scanner system; implemented complete firmware stack in C/C++.

PROJECTS

LoRaNet Communication Module | *KiCad, ESP32, LoRa, C/C++*

- Designed full schematic and multi-layer PCB for a long-range wireless system achieving ~7 km LoS range.
- Implemented embedded firmware for reliable packet transmission over LoRa transceivers.

DC–DC Buck Converter | *KiCad, XL2013E1, Power Electronics*

- Designed 12V-to-5V converter from schematic through PCB; iterated layout to achieve ~90% efficiency and reduced ripple suitable for embedded/industrial use.

Micromouse Autonomous Maze-Solving Robot | *Custom PCB, PID, QTR Sensors*

- Designed custom PCB reducing board size by 40%; achieved 95% navigation accuracy with PID motion control.
- 1st Place** LOCUS 2025 (IOE Pulchowk), **2nd Place** Delta 5.0 (IOE Purwanchal), **5th Place** Techfest IIT Bombay.

Pocket CubeSat Module (1.5U PocketQube) | *Stacked PCB, LoRa, NodeMCU*

- Designed stacked PCB architecture integrating OBC, power management, sensors, and LoRa comms in a 1.5U form factor with Li-Po solar charging.

Solar Greenhouse Automation System | *ESP32, BLE, MOSFET, OLED*

- Built solar-powered system monitoring soil, temperature, and humidity; automated irrigation via MOSFET control with BLE data transmission and real-time OLED dashboard.

Smart Helmet Safety System | *IoT, BLE, SpO2, Fall Detection*

- Developed wearable prototype with pulse/SpO2 sensing, fall detection, and SOS alerts transmitted via BLE to a supervisor dashboard.

EDUCATION

Pashchimanchal Campus (Tribhuvan University)

B.E. in Electronics, Communication & Information Engineering

Pokhara, Nepal

2023 – Present (4th Year)

LEADERSHIP

Electronics & Software Lead

2024 – 2026

Robotics Club, Pashchimanchal Campus

- Mentored 350+ students in embedded systems and PCB design; initiated in-house PCB prototyping lab.
- Led competition robotics projects; organized workshops and technical training programs.

ACHIEVEMENTS

Top 25 Nationwide – U.S.–Nepal Startup Weekend Challenge (Safety Modules), 2026

International Finalist – Techfest IIT Bombay, representing Nepal (Safety Modules), 2025

1st Place – LOCUS 2025 Micromouse, IOE Pulchowk Campus

1st Runner-Up – Autonomous Boat Racing, Robotics Club WRC 2024

1st Place – Line Follower Competition, Robotics Club WRC 2023